HARSHA GURNANI, PhD

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PROFESSIONAL APPOINTMENTS

Faculty Fellow, Center for Data Science, *New York University*.		2026-
Postdoctoral researcher, *New York University*.	Advisor: Prof Cristina Savin	2025-2026
Schmidt Science Fellow, *University of Washington*.	Advisor: Prof Bing Brunton	2023-2025
Undergraduate research intern, *Brandeis University*.	Advisor: Prof Eve Marder, Dr Tim O'Leary	2015-2016

EDUCATION

PhD in Neuroscience, *University College London* Advisor: Prof R Angus Silver		2016-2022
Bachelor of Science with Major in **Biology**, Minor in **Mathematics** *Indian Institute of Science*	[First Class with Distinction]	2012-2016

AWARDS, HONORS, & DISTINCTIONS

NYU CDS Faculty Fellowship		2026-2028
Schmidt Catalyst Grant (joint with Juncal Arbelaiz)		2024-2026
UW/eScience Data Science Fellowship		2023-2025
Schmidt Science Fellowship		2022-2024
Wellcome Trust PhD Studentship		2016-2021
Khorana Scholarship for 3-month research internship	[Khorana Scholars Program, DBT]	2015
KVPY 4-year Undergraduate Fellowship	[Dept. of Science and Technology, India]	2012-2016
eScience Postdoctoral Grant		2024
Travel grant from Gatsby Foundation for ICTP Summer Workshop		2024
Travel grant from Computational and Systems Neuroscience (Cosyne 2019)		2019
Travel grant from DeepMind for Cajal Course in Computational Neuroscience (CCCN 2017)		2017
IISc UG University Gold Medal		2016

PUBLICATIONS

A cortico-striatal circuit updates subjective beliefs about latent task states.
De*Maegd ML*, Hocker D,* ***Gurnani H****, Adler-Wachter M, Schindler J, Schiereck SS, Savin C, Constantinople CM*#. bioRxiv — 2026

The NeuroML ecosystem for standardized multi-scale modelling in neuroscience.
Sinha A, Gleeson P*#, Marin B, Dura-Bernal S, Panagiotou S, Crook S, Cantarelli M, Cannon RC, Davison AP,* ***Gurnani H****, Silver RA*#. eLife — 2025

Feedback control of recurrent circuits imposes dynamical constraints on learning.
Gurnani H**#, Liu W, Brunton BW*. eLife — 2024

Accessible interview practices for disabled scientists and engineers.
Greene SM#, Schachat SR*, Arita-Merino N, Cao XE,* ***Gurnani H****, Heyns M, Cagigas ML, Maikawa CL, Needham EJ, Perets EA, Phillips E, Waddle AW, Wilkinson CE, Zhou KC, Zlotnick HM*. iScience — 2024

Signatures of task learning in neural representations.
Gurnani H**, Cayco-Gajic NA*#. Curr Opinion in Neurobiology (Computational Neuroscience) — 2023

Multidimensional population activity in an electrically coupled inhibitory circuit in the cerebellar cortex.
Gurnani H**, Silver RA*#. Neuron — 2021

Cerebellar granule cell axons support high dimensional representations.
Lanore F, Cayco-Gajic NA*,* ***Gurnani H****, Coyle D, Silver RA*#. Nature Neuroscience — 2021

Dopaminergic and Prefrontal Basis of Learning from Sensory Confidence and Reward Value.
Lak A#, Okun M, Moss MM,* ***Gurnani H,*** *Farrell K, Wells MJ, Reddy CB, Kepecs A, Harris KD, Carandini M.* Neuron — 2020

* first author(s), # corresponding author(s)

TEACHING AND MENTORING

Sensation, Perception and Beyond, UW Bothell;	Co-instructor and course developer	2024
STEP-WISE Scholar, University of Washington;	Pedagogical training	2023-2024
BIOL0029: Computational Biology, UCL;	Teaching Assistant (TA)	2021
Data Science and Machine Learning in Python, UCL;	Course instructor and content developer	2020, 2021
NEUR0019: Neuroinformatics (Methods in Quantitative Neurophysiology), UCL;	TA	2020, 2021
Systems Training in Maths, Informatics, Statistics and Computational Biology, UCL;	TA	2018, 2019
Computational Approaches to Memory and Plasticity (CAMP), Bangalore;	TA	2016

Mentees:

Sierra Caseus (Shenoy Undergraduate Research Fellow, Simons Foundation)	2026-
Chenxiao Yang (undergraduate research intern, Tsinghua University)	2026-
Yiwen Xu (undergraduate research intern, NYU Shanghai)	2025
Jessica Schmilovich (high school student, NYU GSTEM program)	2025
Jianqiao (Lawrence) Hu (Neuroscience PhD student, UW)	2023-2025
Weixuan Liu (undergraduate researcher, UW)	2023-2025
Brennan Summy (undergraduate researcher with ENDURE program, UW)	2024
David Orme (MRC PhD student, UCL)	2020-21
Lewis Winyard (undergraduate thesis project, UCL)	2019-20
Grade 5/6 students with ReachOut, UK	2019-2021

INVITED TALKS

National Centre for Biological Sciences, India	Feb 2026
Indian Institute of Science, Bangalore, India	Feb 2026
Junior Theoretical Neuroscientist Workshop, Flatiron Institute	July 2025
Cosyne 2025 – Workshop on “Dynamics of brain computations through the lens of control theory”	Mar 2025
Gatsby Unit, University College London	Nov 2024
Bernstein Conference – Workshop on “Bridging RNNs and Data”	Sep 2024
Orsborn lab, University of Washington	July 2024
NCEC – Neural Computation and Engineering Connection, UW	May 2024
5th France Cerebellar Meeting, Bordeaux, France	Feb 2022
Janelia Junior Scientist Workshop on Theoretical Neuroscience	Oct 2019

OTHER PRESENTATIONS

Project:	**Dynamics-based alignment across sessions reveals latent neural computation**	
	Harsha Gurnani*, Christine Constantinople, Cristina Savin	
[Poster]	Gordon Research Conference (GRC) – Cognition	July 2026
[Poster]	Cosyne 2026	Mar 2026
[Talk]	Center for Neural Science, NYU	Nov 2025

Project:	**Feedback control of recurrent circuits imposes dynamical constraints on learning**	
	Harsha Gurnani*, *BW Brunton*	
[Talk]	Junior Theoretical Neuroscientist Workshop, Flatiron Institute	July 2025
[Poster]	Data-Driven Discovery: AI and Modelling in Biology (Allen Institute)	Sep 2024
[Talk]	ICTP Workshop on Recent Advances in Theoretical Neuroscience	Jun 2024
[Talk]	Mathematics Of Neuroscience and AI	May 2024
[Poster]	Cosyne 2024	Feb 2024

Project:	**Transformation of cortico-pontine inputs during associative learning**	
	Harsha Gurnani*, *RA Silver*	
[Poster]	Lake Conference, Neural Dynamics	Oct 2023

Project:	**Cerebellar-like structure improves feedback-learning in recurrent neural networks**	
	Alessandro Barri,* ***Harsha Gurnani****, RA Silver*	
[Poster]	ICTP Workshop on Recent Advances in Theoretical Neuroscience	Jun 2024

[Poster]	Cosyne 2021	Feb 2021

Project: **Dynamics of electrically coupled inhibitory networks in the cerebellar cortex**

Harsha Gurnani*, *RA Silver*

[Poster]	Cosyne 2021	Feb 2021
[Talk]	Gordon Research Seminar & Conference (GRC Cerebellum)	July 2019
[Poster]	Computational and Systems Neuroscience conference (Cosyne 2019)	Mar 2019
[Poster]	3rd France Cerebellar Meeting	Jan 2019
[Poster]	Champalimaud Research Symposium	Oct 2018

Project: **Imaging circuit function across multiple scales with non-linear acousto-optic microscopy**

RA Silver, Antoine Valera, ***Harsha Gurnani***, *T J Younts, VA Griffiths, S Punde, TF Alfonso, P A Kirkby, KMNS Nadella*

[Poster]	NIH Brain Initiative meeting	Jun 2021

Project: **Dopaminergic and frontal signals for reward learning in perceptual decisions**

Armin Lak, M Okun, M Moss,* ***Harsha Gurnani****, MJ Wells, CB Reddy, KD Harris, M Carandini*

[Poster]	Neuroscience 2018 (SfN)	Nov 2018
[Poster]	Neuroscience 2017 (SfN)	Nov 2017

Project: **Maintaining neuronal properties during growth with local and global homeostatic regulation**

Harsha Gurnani**, T O' Leary, E Marder*

[Poster]	Neuroscience 2016 (SfN)	Nov 2016
[Talk]	Dynamic Neural Networks: STG Meeting 2015	Oct 2015

WORKSHOPS AND SCHOOLS

Control Theory and Neuroscience, Flatiron Institute	May 2026
Junior Theoretical Neuroscience Workshop, Flatiron Institute	July 2025
Data-Driven Discovery: AI and Modeling in Biology	Sep 2024
ICTP Junior Scientist Workshop on Advances in Theoretical Neuroscience	Jun 2024
Janelia Junior Scientist Workshop on Theoretical Neuroscience	Oct 2019
Optical Imaging and Electrophysiological Methods in Neuroscience	May 2018
CAJAL Course in Computational Neuroscience	Aug 2017
Computational Approaches to Memory and Plasticity	July 2014
Bangalore Cognition Workshop	Dec 2013

VOLUNTEERING AND SERVICE

Writer/Editor/Interviewer at Stories of Women in Neuroscience (*Stories of WiN*)	2025-
Cosyne workshop ("*Learning fast and slow*"); co-organized with Jacob Sacks, Matt Golub	2025
Organizing committee, UCL NeuroAI	2022
Organizing committee, UCL PhDs in Systems Neuroscience	2019-2020
PhD student committee, NPP, UCL	2018-2019
Winter volunteer with CRISIS UK	2016-2018
Volunteer at Notebook Drive, IISc	2013-2016
Ad hoc reviewer for *Science*, *Cell*, *Neuron*, *Journal of Neuroscience*, *PLOS CB*, *Nature Neuroscience*, *Cosyne*, Schmidt Polymaths Program, *UW Biology* departmental awards	2019-